
EDUCATION

Stanford University

Stanford, CA

B.S. **Mechanical Engineering**; Graduating in June 2027; GPA: **3.812**

Sep 2023 – Present

Relevant coursework: ENGR 14: Introduction to solid mechanics, ME 30: Thermodynamics, ME 70: Fluid dynamics, ME 80: Mechanics of Materials, ME 149: Mechanical Measurements, ME 102: Foundations of Product Realization, ENGR 40 M, ENGR 15: Dynamics, AA126: Aerospace Prototyping Lab

SKILLS

- **Programming & Controls:** Python, Arduino, MATLAB, JavaScript, C++
- **CAD & Simulation:** Fusion360, Onshape, SolidWorks, CNC, Luminary Cloud, ANSYS Fluent, HyperMesh, OptiStruct
- **Fabrication:** CNC machining, 3D printing, laser cutting, soldering, sand casting, welding

WORK EXPERIENCES

Stanford Shape Lab

Stanford, CA

Undergraduate Research Assistant

Dec 2023 – Present

Stanford Mechanical Engineering Department laboratory focused on the development of cutting-edge robotics, mechatronics, and sensing technologies with the primary goal to create dynamic 3D displays and haptic interfaces

<https://shape.stanford.edu/>

- Prototyped multiple wearable haptic-feedback devices, integrating LRAs, vibration motors, Dayton Audio exciters, DRV2605L drivers, amplifiers, and a TCA9548A I²C multiplexer into compact wristband designs.
- Worked through multi-channel hardware issues including power drops, signal noise, I²C address conflicts, motor placement, vibration strength, and wiring.
- Set up the Arduino and Node.js backend that connects the wristband to the web app, letting it send real-time vibration feedback based on handwriting errors.
- Ran hardware validation and pilot studies to measure latency, feedback precision, and user perception across prototype iterations.

Stanford Solar Car Project

Stanford, CA

Member of Aero Sub team, Mechanical Division

Mar 2025 – Present

Student-run organization at Stanford University dedicated to designing and building solar-powered cars including work on the most recent entry in the 2025 Formula Sun Grand Prix in Kentucky (2nd place)

- Designed and surfaced aero body geometry with manufacturable, watertight CAD, improving curvature continuity, leading-edge shaping, and canopy integration.
- Ran aerodynamic simulations evaluating drag, separation behaviour, and crosswind/yaw stability.
- Built the CFD workflow for meshing, solver setup, and analysis of pressure distribution and wake structure.
- Analyzed nose geometry, canopy location, cooling-airflow behavior, and underbody laminar flow to reduce drag and improve stability.

PROJECTS

Mechanical Design Project

Apr – Jun 2025

ME 102, Stanford University

- Built a lever-actuated escapement and multi-stage gear train (planetary, bevel, worm-wheel). Performed kinematic, stress, and tolerance analyses. Fabricated via 3D printing in PLA and laser cutting.

Mechanical Measurement Labs

Apr – Jun 2025

ME 149, Stanford University

- Conducted rheology studies of non-Newtonian fluids, tensile testing of polymers, heat transfer experiments (Biot numbers, convection coefficients), and PIV vortex-ring analysis.

Wing Test Project

Sep – Nov 2025

AA126, Stanford University

- Designed and fabricated an airfoil wing section with custom mounting hardware, then calibrated the load-cell system and ran wind-tunnel tests across multiple angles of attack to analyze lift and drag performance.

AA126: Wind Tunnel Project

Sep 2025 – present

AA126, Stanford University

- Constructing a wind-tunnel assembly by CNC-cutting and heat-forming acrylic panels to build the contraction, test-section, and diffuser geometry, integrating a honeycomb flow straightener to condition the inlet stream and ensure uniform velocity profiles.

